Saif Mohammed

Systems and Biomedical Engineering Student

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Education

Cairo University – Faculty of Engineering *Expected Graduation: July 2027*
B.Sc. in Systems and Biomedical Engineering
Relevant Coursework: Deep Learning, Linear Algebra, Multivariate Calculus, Probability and Statistics, Object-Oriented Programming, Data Structures and Algorithms

Experience

- **Undergraduate Researcher, RiSH** *April 2026 – Present*
  Advised by **Dr. Jamil Fayyad**
  - Researching conformal prediction within world models in medical applications.
  - Co-authored an under-review paper on Conformalizing Medical World Models through Predictor Differentials.
  - My contribution involved running the initial experiments to test our hypothesis about how the world models behave under distribution shift.

- **Undergraduate Research Assistant, Cairo University** GitHub *December 2025 – April 2026*
  Advised by **Dr. Meena M. Makary**
  - Developing a longitudinal Multiple Sclerosis lesion tracking and labeling system for multi-timepoint MRI data.
  - Implementing medical image processing and registration pipelines using NumPy, SciPy, ANTsPy and NiBabel.

- **AI Developer Intern, Siemens DISW** *July 2025 – August 2025*
  - Designed and implemented LLM-powered features for internal engineering tools using existing LLM APIs and Retrieval-Augmented Generation (RAG) systems.
  - Built structured prompt pipelines to improve response relevance and reliability.

Technical Skills

- **Programming:** Python, C, C++, Bash
- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, NiBabel
- **Backend & Deployment:** Flask, FastAPI, Docker, CI/CD
- **Tools:** Git, GitHub, Linux, LaTeX, OpenCode, Claude Code

Projects

Ego Curation *(Graduation Project, Ongoing)* GitHub | *Deep Learning, Video dataset, Egocentric dataset, World Models*
- Building a pipeline to curate an egocentric video dataset (Ego4D) by extracting the most surprising segments, reducing dataset size for training a sample-efficient world model with improved physics intuition.
- Using state-of-the-art video world models (V-JEPA 2 & 2.1) as a baseline for the surprise metric calculation.

RobustNet GitHub | *Deep Learning, CNN, Conformal Prediction, Robust AI*
- Trained a ResNet32 network from scratch for 186 epochs, reproducing the original paper test error rate of 7.6%.
- Applied training best practices including early stopping, LR scheduling, and frequent checkpoints.
- Followed PyTorch performance tuning guide for best practices to accelerate training and inference.
- Integrated a conformal prediction layer to improve model robustness and calibration, and to detect out-of-distribution examples.

Seq2Seq Model From Scratch GitHub | *Deep Learning, Sequence Models, LSTM*
- Implemented an end-to-end Seq2Seq architecture based on Sutskever et al.
- Built an English-French neural machine translation system trained on 34,802 sentence pairs.
- Achieved a 38.3 BLEU score with 173 sentences per second inference speed.

Honors & Awards

- **NASA Space Apps Cairo Local Winner & Global Nominee** *Oct 2024*
- **4th Place - Undergraduate Engineering Mathematics Research Forum** *Dec 2024*
  Cairo University - Technical Center for Career Development

Extracurricular

- **Head of AI Committee - IEEE EMBS CUSB**
- **Autonomous Team Member - Cairo University Racking Team**
- **English Instructor - CUFE in English**